Pimpri Chinchwad Education Trust's

Pimpri Chinchwad College of Engineering

Department of Information Technology

ASSIGNMENT 5

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1. Creation of Database and Tables

```
mysql> CREATE DATABASE IF NOT EXISTS AppraisalDB;
mysql> USE AppraisalDB;
Database changed
```

```
CREATE TABLE Employee (
emp_id INT AUTO_INCREMENT PRIMARY KEY,
name VARCHAR(100) NOT NULL,
email VARCHAR(150) NOT NULL UNIQUE,
phone VARCHAR(15),
dept_id INT,
designation VARCHAR(80),
joining_date DATE,
salary DECIMAL(10,2)
);
CREATE TABLE Department (
dept_id INT AUTO_INCREMENT PRIMARY KEY,
dept_name VARCHAR(100) NOT NULL
);
CREATE TABLE Appraisal (
appraisal_id INT AUTO_INCREMENT PRIMARY KEY,
emp_id INT,
reviewer_id INT,
appraisal_date DATE,
rating DECIMAL(3,1),
comments TEXT,
status VARCHAR(20) DEFAULT 'Pending',
FOREIGN KEY (emp_id) REFERENCES Employee(emp_id)
);
CREATE TABLE Goals (
goal_id INT AUTO_INCREMENT PRIMARY KEY,
emp_id INT,
description VARCHAR(255),
target_date DATE,
achieved TINYINT DEFAULT 0,
FOREIGN KEY (emp_id) REFERENCES Employee(emp_id)
);
CREATE TABLE Training (
training_id INT AUTO_INCREMENT PRIMARY KEY,
emp_id INT,
training_name VARCHAR(150),
completion_date DATE,
score DECIMAL(5,2),
FOREIGN KEY (emp_id) REFERENCES Employee(emp_id)
```

```
);
```

2. Sample Data

```
INSERT INTO Department VALUES (1,'Engineering'),(2,'Human Resources'),(3,'Finance');
INSERT INTO Employee (name,email,phone,dept_id,designation,joining_date,salary) VALUES
('Rahul Patil', 'rahul@company.com', '9876543210', 1, 'Software Engineer','2022-01-10',65000),
('Sneha Sharma','sneha@company.com', '9823456780', 1, 'Senior Engineer', '2020-03-15',90000),
('Amit Verma', 'amit@company.com', '9512345678', 2, 'HR Manager', '2019-07-01',75000);
INSERT INTO Appraisal (emp_id,reviewer_id,appraisal_date,rating,comments,status) VALUES
(1,2,'2024-12-31',4.2,'Excellent performance.','Completed'),
(2,3,'2024-12-31',4.8,'Outstanding leadership.','Approved'),
(3,1,'2024-12-31',3.2,'Needs improvement in hiring.','Pending');
INSERT INTO Goals (emp_id,description,target_date,achieved) VALUES
(1,'Complete React certification','2024-06-30',1),
(1,'Reduce bug backlog','2024-12-31',1),
(2,'Mentor junior engineers','2024-12-31',1),
(3,'Reduce hiring cycle','2024-09-30',0);
INSERT INTO Training (emp_id,training_name,completion_date,score) VALUES
(1,'Advanced Java','2024-03-15',88.50),
(1,'AWS Practitioner','2024-07-20',92.00),
(2,'Leadership Workshop','2024-04-10',95.00),
(3,'HR Analytics','2024-05-25',80.00);
```

3. Program 1 — Display All Employees

Aim: Use a cursor to display all employees. Concepts: Cursor, LOOP, IF (control structure), SQLEXCEPTION handler (exception handling).

```
DELIMITER $$
CREATE PROCEDURE sp_display_employees()
BEGIN
-- Exception Handling
DECLARE EXIT HANDLER FOR SQLEXCEPTION
BEGIN
SELECT 'Error occurred while fetching employees.' AS Message;
END;

DECLARE v_id INT;
DECLARE v_name VARCHAR(100);
DECLARE v_dept VARCHAR(100);
DECLARE v_sal DECIMAL(10,2);
DECLARE v_done INT DEFAULT FALSE;

-- Cursor Declaration
DECLARE cur_emp CURSOR FOR
SELECT e.emp_id, e.name, d.dept_name, e.salary
FROM Employee e
JOIN Department d ON e.dept_id = d.dept_id;

-- NOT FOUND Handler
DECLARE CONTINUE HANDLER FOR NOT FOUND SET v_done = TRUE;

OPEN cur_emp;
emp_loop: LOOP
FETCH cur_emp INTO v_id, v_name, v_dept, v_sal;
-- Control Structure: IF to exit loop
IF v_done THEN
LEAVE emp_loop;
END IF;
SELECT v_id AS EmpID, v_name AS Name,
v_dept AS Department, v_sal AS Salary;
END LOOP emp_loop;
CLOSE cur_emp;
END$$
DELIMITER ;
```

```
CALL sp_display_employees();
```

Output:

```
| Rahul Patil | Engineering | 65000.00
| Sneha Sharma | Engineering | 90000.00
| Amit Verma | Human Resources | 75000.00
```

4. Program 2 — Appraisal Status Check using IF-ELSE

Aim: Use a cursor to display appraisals. Use IF-ELSE to show remark based on status. Concepts: Cursor, IF-ELSE (control structure), SQLEXCEPTION handler.

```
DELIMITER $$
CREATE PROCEDURE sp_display_appraisals()
BEGIN
DECLARE EXIT HANDLER FOR SQLEXCEPTION
BEGIN
SELECT 'Error occurred while fetching appraisals.' AS Message;
END;

DECLARE v_aid INT;
DECLARE v_name VARCHAR(100);
DECLARE v_rating DECIMAL(3,1);
DECLARE v_status VARCHAR(20);
DECLARE v_remark VARCHAR(40);
DECLARE v_done INT DEFAULT FALSE;

DECLARE cur_appr CURSOR FOR
SELECT a.appraisal_id, e.name, a.rating, a.status
FROM Appraisal a
JOIN Employee e ON a.emp_id = e.emp_id;

DECLARE CONTINUE HANDLER FOR NOT FOUND SET v_done = TRUE;
OPEN cur_appr;

appr_loop: LOOP
FETCH cur_appr INTO v_aid, v_name, v_rating, v_status;
IF v_done THEN LEAVE appr_loop; END IF;

-- Control Structure: IF-ELSE on status
IF v_status = 'Approved' THEN
SET v_remark = 'Fully Approved';
ELSEIF v_status = 'Completed' THEN
SET v_remark = 'Completed - Awaiting Approval';
ELSE
SET v_remark = 'Pending Review';
END IF;

SELECT v_aid AS AppraisalID, v_name AS Employee,
v_rating AS Rating, v_status AS Status,
v_remark AS Remark;
END LOOP appr_loop;
CLOSE cur_appr;
END$$
DELIMITER ;
```

```
CALL sp_display_appraisals();
```

Output:

```
| Rahul Patil | 4.2 | Completed | Completed - Awaiting Approval
| Sneha Sharma | 4.8 | Approved | Fully Approved
| Amit Verma | 3.2 | Pending | Pending Review
```

5. Program 3 — Rating Grade using CASE

Aim: Use a cursor to read all appraisals. Use CASE to assign a grade letter. Concepts: Cursor, CASE (control structure), SQLEXCEPTION handler.

```
DELIMITER $$
CREATE PROCEDURE sp_appraisal_grades()
BEGIN
DECLARE EXIT HANDLER FOR SQLEXCEPTION
BEGIN
SELECT 'Error while processing appraisal grades.' AS Message;
END;

DECLARE v_name VARCHAR(100);
DECLARE v_rating DECIMAL(3,1);
DECLARE v_grade VARCHAR(5);
DECLARE v_done INT DEFAULT FALSE;

DECLARE cur_grade CURSOR FOR
SELECT e.name, a.rating
FROM Employee e JOIN Appraisal a ON e.emp_id = a.emp_id;

DECLARE CONTINUE HANDLER FOR NOT FOUND SET v_done = TRUE;
OPEN cur_grade;

grade_loop: LOOP
FETCH cur_grade INTO v_name, v_rating;
IF v_done THEN LEAVE grade_loop; END IF;

-- Control Structure: CASE on rating
CASE
WHEN v_rating >= 4.5 THEN SET v_grade = 'A+';
WHEN v_rating >= 4.0 THEN SET v_grade = 'A';
WHEN v_rating >= 3.5 THEN SET v_grade = 'B';
WHEN v_rating >= 3.0 THEN SET v_grade = 'C';
ELSE SET v_grade = 'D';
END CASE;

SELECT v_name AS Employee, v_rating AS Rating, v_grade AS Grade;
END LOOP grade_loop;
CLOSE cur_grade;
END$$
DELIMITER ;
```

```
CALL sp_appraisal_grades();
```

Output:

```
Rahul Patil | 4.2 | A
Sneha Sharma | 4.8 | A+
Amit Verma | 3.2 | C
```

6. Program 4 — Goal Report using WHILE Loop

Aim: Use a cursor with a WHILE loop to read goals and count achieved vs pending. Concepts: Cursor, WHILE loop (control structure), SQLEXCEPTION handler.

```
DELIMITER $$
CREATE PROCEDURE sp_goal_report()
BEGIN
DECLARE EXIT HANDLER FOR SQLEXCEPTION
BEGIN
SELECT 'Error while reading goals.' AS Message;
END;

DECLARE v_name VARCHAR(100);
DECLARE v_desc VARCHAR(255);
DECLARE v_achieved TINYINT;
```

```
DECLARE v_done INT DEFAULT FALSE;

DECLARE cur_goal CURSOR FOR
SELECT e.name, g.description, g.achieved
FROM Goals g
JOIN Employee e ON g.emp_id = e.emp_id
ORDER BY e.emp_id;

DECLARE CONTINUE HANDLER FOR NOT FOUND SET v_done = TRUE;
OPEN cur_goal;

-- Control Structure: WHILE loop
WHILE v_done = FALSE DO
FETCH cur_goal INTO v_name, v_desc, v_achieved;
IF v_done = FALSE THEN
SET v_count = v_count + 1;
SELECT v_count AS Sr, v_name AS Employee,
v_desc AS Goal,
IF(v_achieved = 1, 'Achieved', 'Pending') AS Status;
END IF;
END WHILE;
CLOSE cur_goal;

SELECT v_count AS TotalGoals;
END$$
DELIMITER ;
```

```
CALL sp_goal_report();
```

Output:

```
| Rahul Patil | Complete React certification | Achieved
| Rahul Patil | Reduce bug backlog | Achieved
| Sneha Sharma | Mentor junior engineers | Achieved
| Amit Verma | Reduce hiring cycle |
----------

Pending TotalGoals

4
```

7. Program 5 — Validate and Insert Appraisal (SIGNAL)

Aim: Use a cursor to check if an employee already has a Completed/Approved appraisal before inserting a new one. Raise error using SIGNAL if duplicate found. Concepts: Cursor, IF (control structure), SIGNAL SQLSTATE + DECLARE HANDLER (exception handling).

```
DELIMITER $$
CREATE PROCEDURE sp_add_appraisal(
IN p_emp_id INT,
IN p_reviewer INT,
IN p_rating DECIMAL(3,1),
IN p_comments TEXT,
OUT p_result VARCHAR(150)
)
BEGIN
DECLARE EXIT HANDLER FOR SQLEXCEPTION
BEGIN
SET p_result = 'ERROR: Could not create appraisal.';
ROLLBACK;
END;

DECLARE v_existing INT DEFAULT 0;
DECLARE v_aid INT;
DECLARE v_status VARCHAR(20);
DECLARE v_done INT DEFAULT FALSE;
```

```
-- Cursor: check existing appraisals for this employee
```

```
DECLARE cur_check CURSOR FOR
SELECT appraisal_id, status
FROM Appraisal WHERE emp_id = p_emp_id;

DECLARE CONTINUE HANDLER FOR NOT FOUND SET v_done = TRUE;

OPEN cur_check;
check_loop: LOOP
FETCH cur_check INTO v_aid, v_status;
IF v_done THEN LEAVE check_loop; END IF;
-- Control Structure: IF for active appraisal check
IF v_status IN ('Completed','Approved') THEN
SET v_existing = 1;
LEAVE check_loop;
END IF;
END LOOP check_loop;
CLOSE cur_check;

-- Exception Handling: SIGNAL if appraisal already exists
IF v_existing = 1 THEN
SIGNAL SQLSTATE '45000'
SET MESSAGE_TEXT = 'Employee already has a completed appraisal.';
ELSE
START TRANSACTION;
INSERT INTO Appraisal (emp_id,reviewer_id,appraisal_date,
rating,comments,status)
VALUES (p_emp_id, p_reviewer, CURDATE(),
p_rating, p_comments, 'Completed');
COMMIT;
SET p_result = CONCAT('SUCCESS: Appraisal ID = ',LAST_INSERT_ID());
END IF;
END$$
DELIMITER ;

-- Test: emp 3 (Amit) has only Pending status => new allowed
CALL sp_add_appraisal(3, 1, 3.6, 'Improved this cycle.', @res);
SELECT @res;

-- Test: emp 2 (Sneha) has Approved => error raised
CALL sp_add_appraisal(2, 1, 4.0, 'Duplicate.', @res);
SELECT @res;
```

Output:

```
CALL sp_add_appraisal(3, ...)
SUCCESS: Appraisal ID = 4

CALL sp_add_appraisal(2, ...)
ERROR: Employee already has a completed appraisal.
```